

Batch: A3 Roll No.: 16010324044
Experiment / assignment / tutorial No. 6
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: MOSFET Biasing

AIM: To analyse, design fixed bias and self-bias circuits of MOSFET

1. **OUTCOME:** Student will be able to analyze the transistor circuits for DC operation

I) Design of Fixed Bias MOSFET

Design Fixed bias MOSFET circuit for the given conditions- $I_{DQ} = 3 \text{ mA}$, $V_{DSQ} = 5 \text{ V}$. Assume suitable value of V_{DD} . Given that $\mu_n C_{ox} = 100 \mu\text{A/V}^2$, $(W/L) = 6/0.1$, $V_{TN} = 0.5 \text{ V}$, $\lambda = 0$

① $I_{DQ} = 3 \text{ mA}$, $V_{DSQ} = 5 \text{ V}$, $V_{DD} = 10 \text{ V}$, $\mu_n C_{ox} = 100 \mu\text{A/V}^2$, $(\frac{W}{L}) = \frac{6}{0.1}$, $V_{TN} = 0.5 \text{ V}$, $\lambda = 0$

To calculate V_{GS}

$$I_D = \frac{\mu_n C_{ox}}{2} \left(\frac{W}{L}\right) (V_{GS} - V_{TN})^2$$

$$\rightarrow 3 \times 10^{-3} = \frac{100 \times 10^{-6}}{2} \left(\frac{6}{0.1}\right) (V_{GS} - 0.5)^2$$

$$\rightarrow (V_{GS} - 0.5) = \pm 1$$

$$\rightarrow V_{GS} = 1.5 \text{ V} \text{ or } V_{GS} = -0.5 \text{ V}$$

(only +ve value)

To calculate R_D (Drain Resistance)

$$\therefore V_{DD} = 10 \text{ V}$$

$$\therefore V_{DD} = V_{DSQ} + I_D R_D$$

$$\rightarrow 10 = 5 + (3 \times 10^{-3}) R_D$$

$$\rightarrow 5 = (3 \times 10^{-3}) R_D$$

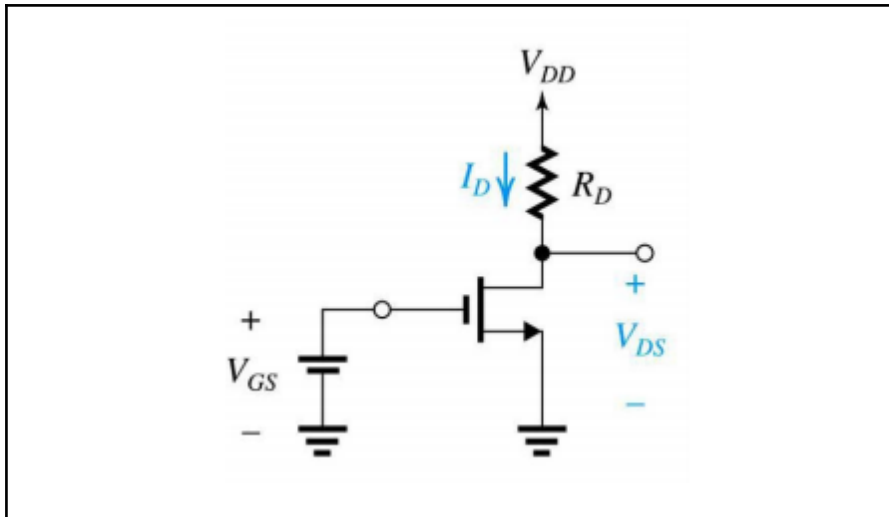
$$\rightarrow R_D = \frac{5}{3 \times 10^{-3}} \rightarrow R_D = 1.67 \text{ k}\Omega$$

Procedure:

1. Implement the circuit as per the circuit diagram in workspace of LTspice.
2. Note the observations as per observation table.
3. Comment on region of operation. Determine the value of V_{GS} (transition point) at which the transistor changes the region of operation.

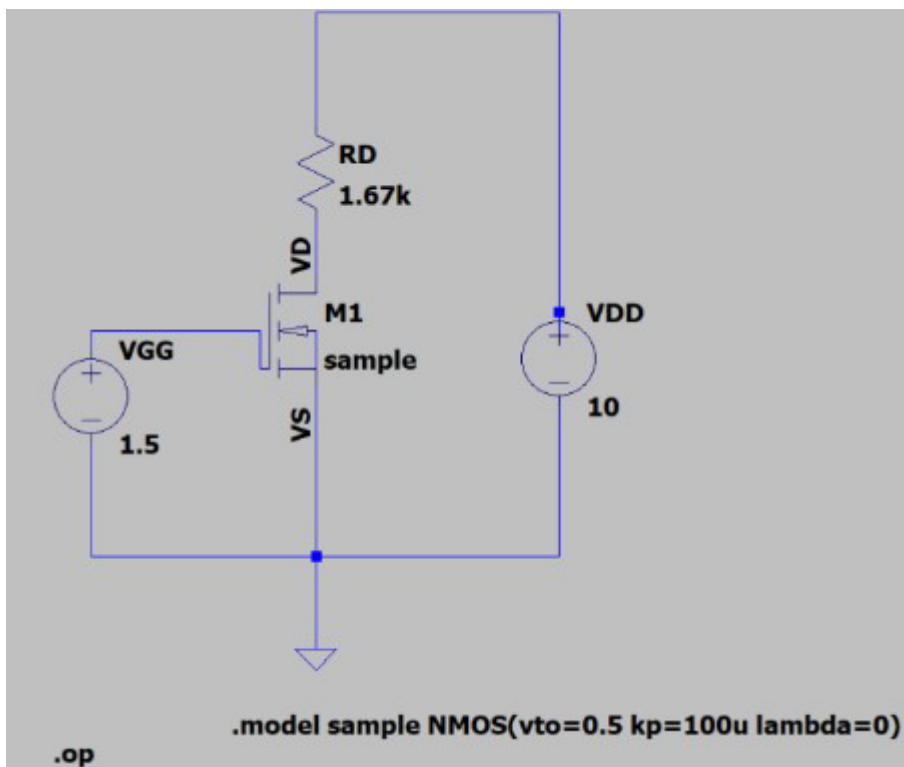
Circuit Diagram:

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Design Steps:-

1. Identify Given Parameters:
2. Assume Suitable VDD
3. Calculate VGS
4. Implement the Circuit in LTspice.
5. Determine Transition Point of VGS



Observation Table:-

	V_{GS}	I_{DQ}	V_{DSQ}
Practically	1.5V	3mA	4.99V

Theoretically	1.5V	3mA	5V
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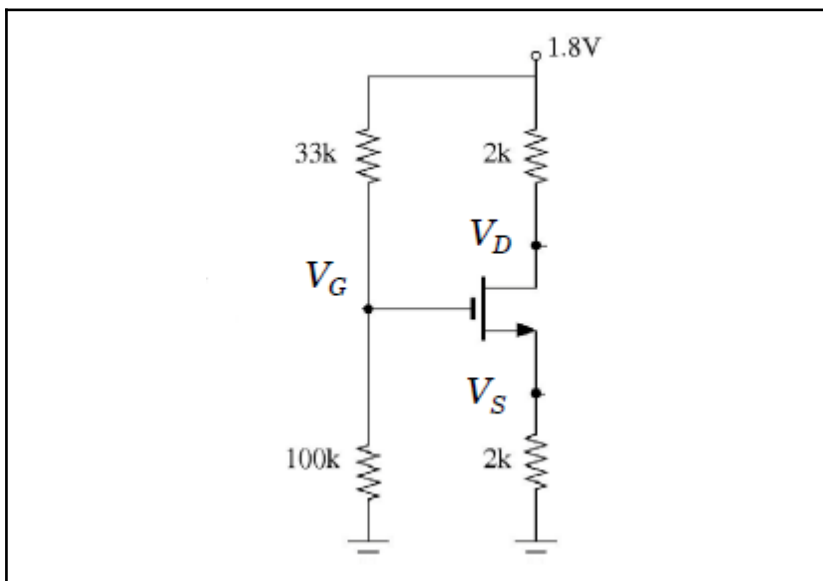
II) **Analysis of Voltage divider bias MOSFET:**

Analyze the given circuit. Given that $\mu nC_{ox} = 100 \mu\text{A}/\text{V}^2$, $(W/L) = 6/0.1$, $V_{TN} = 0.5 \text{ V}$, $V_A = 1 / \lambda = 10 \text{ V}$.

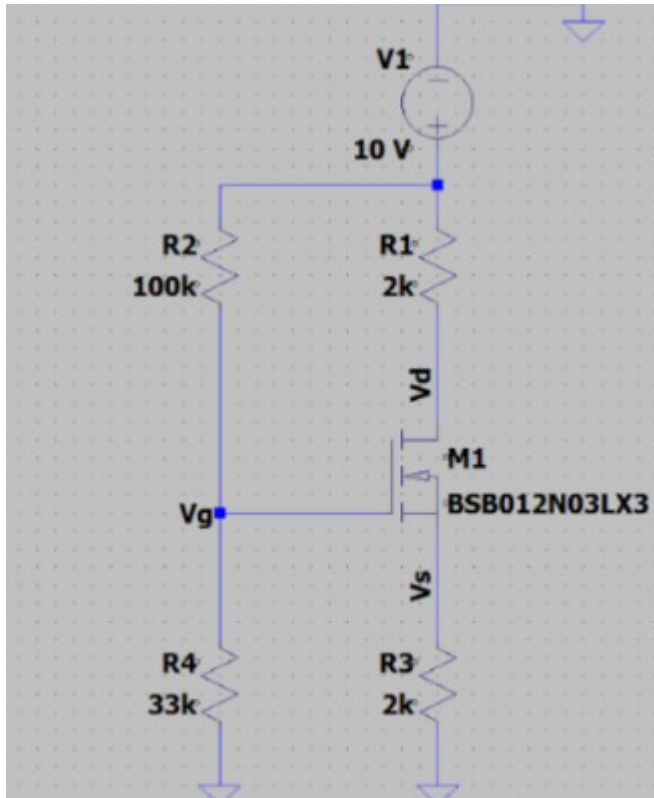
Procedure:

1. Implement the circuit as per the circuit diagram in workspace of LTSpice.
2. Note the observations as per observation table.
3. Comment on the region of operation of MOSFET.

Circuit Diagram:



Circuit Diagram (LTSpice):



Observation Table:-

	V_{GS}	I_{DQ}	V_{DSQ}
Practically	0.919828V	1.14146V	7.71708V
Theoretically	0.19828V	1.14146V	7.71708V

Conclusion:

The experiment successfully demonstrated the design and analysis of fixed and self-bias MOSFET circuits. The observed values of current and voltage were consistent with theoretical predictions, confirming accurate circuit operation.

Signature of Faculty with date